

What your churn data knew, and when it knew it

Prepared for the Board · Period 1 Jul 2024 – 30 Jun 2026

Accounts analysed 4,218 Churn events 1,183 Method Walk-forward, survivorship-safe Issued 8 Aug 2026

Sample report. Figures are synthetic and illustrate the deliverable's structure, not any real company's results. Every number a real report contains is computed from the client's own exported history.

1 VERDICT

Over the last 24 months you lost **\$3.91M of ARR** to churn and a further **\$0.87M** to contraction, against a starting book of \$31.4M, for a gross revenue retention of **84.8%**. Of the churned ARR, **36% was already flaggable 90 days before the cancellation**, and 70% was flaggable at 30 days. Against a random baseline of 11%, that signal is real and well outside chance.

The gap is not detection. It is response. Your median first recorded intervention landed **19 days** before cancellation, deep inside the window where the outcome was largely determined. **\$2.31M of churned ARR, 59% of the total, has no recorded intervention at all.**

\$4.78M	84.8%	36%	19d
ARR lost to churn and contraction	Gross revenue retention	Churned ARR flaggable at 90 days	Median first intervention before churn

2 DATA SUFFICIENCY

Conclusions in this report are drawn only where the underlying data supports them. Cells below the reportability floor are marked and left unquantified rather than printed with confidence intervals too wide to act on.

SOURCE	COVERAGE	EVENT DATING	VERDICT
Billing (Chargebee)	99.2% of accounts	Explicit, to the day	Sufficient
CRM stage history	96.7%	Explicit	Sufficient
Support tickets	88.1%	Explicit	Sufficient
Product usage	41.3%	Monthly rollup only	Partial, excluded

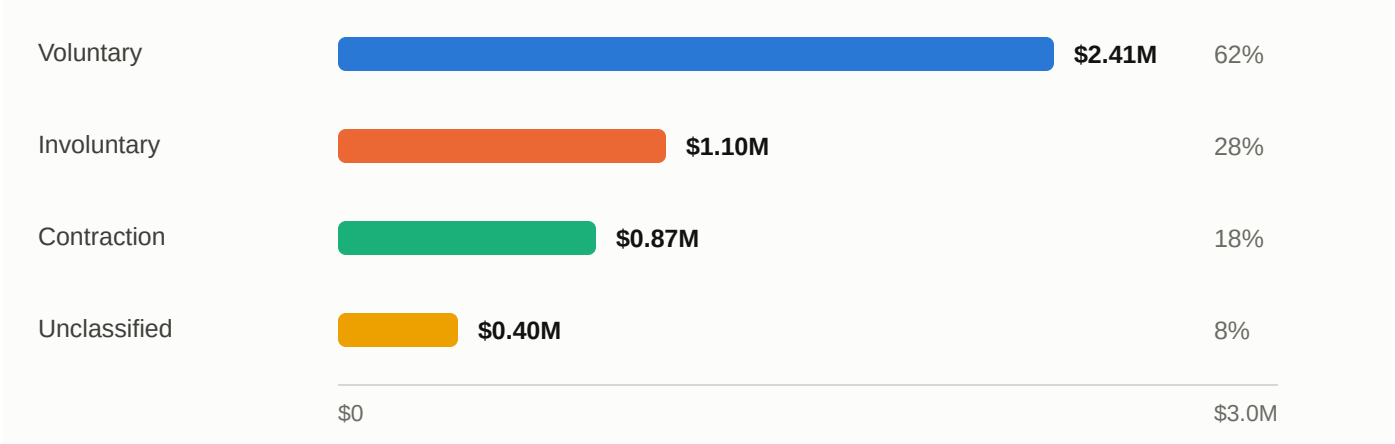
Product usage was excluded from feature reconstruction: monthly rollups cannot be reconstructed to a daily as-of state without leaking information from after the prediction date. Including it would have inflated every accuracy figure in this report.

REPORTABILITY FLOOR

1,183 churn events across a 3-way decomposition and 3 lead-time windows gives 9 cells with a median of **131 events each**, comfortably above the 20-event floor. All cells in this report are reportable. A book of fewer than roughly 1,000 accounts would not have cleared this bar.

3 WHAT YOU LOST

Churn is not one phenomenon. Voluntary cancellation, payment failure, and downgrade have different causes and different fixes, and conflating them produces retention programmes aimed at the wrong third of the problem.



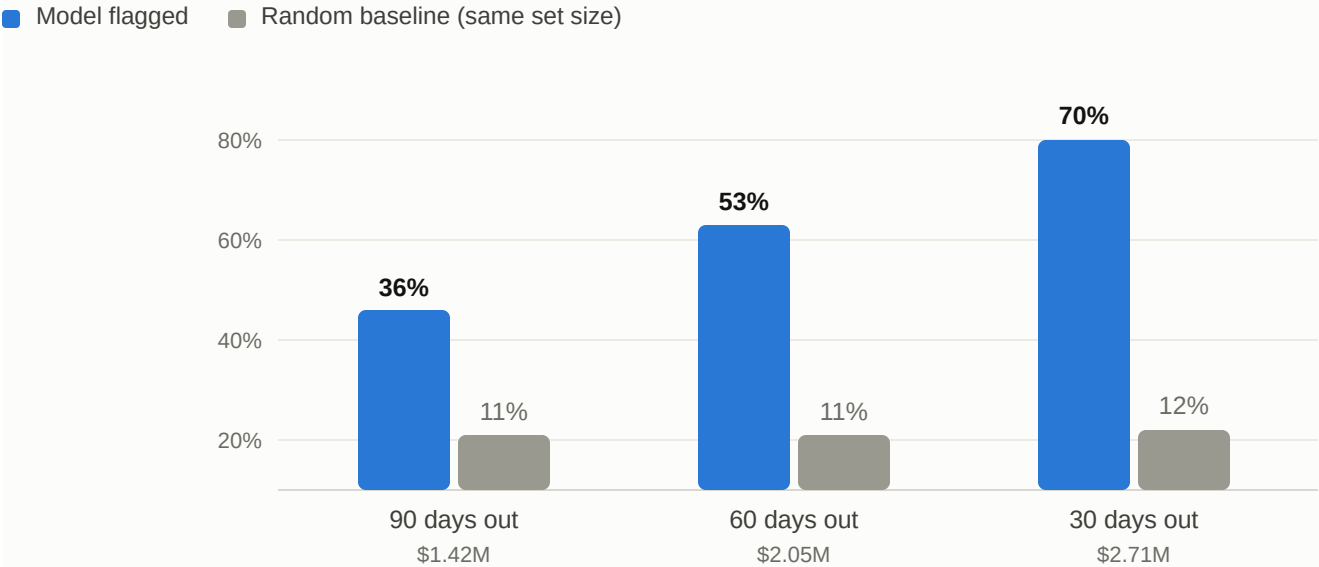
ARR lost by category, 24 months. Percentages are of total ARR lost (\$4.78M) including contraction. Unclassified accounts are those churning without a resolvable ARR value on record.

THE FINDING WORTH ACTING ON FIRST

\$1.10M, or 28% of losses, is involuntary. Failed payments, not decisions to leave. These customers did not choose to go. This is the cheapest retention dollar available to you, and it sits almost entirely outside the customer-success function that is currently accountable for churn.

4 WHAT WAS KNOWABLE IN ADVANCE

For each account that churned, the model was rebuilt using only information that existed at the time. No later data, no survivorship. The bars below show how much of the churned ARR was already in the flagged set at each horizon, beside what a randomly chosen set of the same size would have captured.



Share of churned ARR present in the flagged set at each horizon, against a random set of identical size (20 seeded trials; p95 shown would be 14%, 14% and 15% respectively, and every result clears it). Dollar figures are the flagged ARR at that horizon.

HORIZON	FLAGGED ARR	SHARE	BASELINE	RECALL	PRECISION	EVENTS
90 days	\$1.42M	36%	11%	0.34	0.29	402
60 days	\$2.05M	53%	11%	0.51	0.38	603
30 days	\$2.71M	70%	12%	0.68	0.44	804

5 HOW FAST YOU REACTED

Detection only matters if it is followed by contact. Against the 1,183 accounts that churned, this is what the record shows.

41%

Churned accounts with any recorded intervention

19d

Median days from first intervention to cancellation

\$2.31M

Churned ARR with no recorded intervention

Read alongside Section 4, the shape of the problem is specific: at 90 days the model surfaced \$1.42M of ARR that later churned, but the organisation's median first touch came 71 days later. The detection window was open and largely unused.

READ THIS BEFORE ACTING ON THE NUMBER ABOVE

This is a measure of *recorded* intervention. Conversations held over email, phone or in person without a corresponding CRM record are invisible here, and will inflate the "no intervention" figure. Treat \$2.31M as an upper bound on unattended ARR, not a settled fact.

6 MODEL TRACK RECORD

Twelve walk-forward folds, each trained only on data preceding its origin date, with a 14-day embargo between training and scoring windows to prevent boundary leakage.

METRIC	RESULTINTERPRETATION
Precision @ 20	0.4495% CI 0.38–0.50
Lift over base rate	3.1×Base rate 14.0%
ROC AUC (median fold)	0.71Range 0.66–0.75 across 12 folds
Separable from chance	YesCI excludes the baseline at every horizon
Negative control	0.497Shuffled labels, no residual signal
Leakage canary	CleanNo feature exceeds its as-of availability

The negative control and leakage canary are reported because their failure would invalidate everything above them. A model that scores 0.497 on shuffled labels is finding structure in the data rather than in the evaluation.

7 WHAT THIS REPORT DOES NOT PROVE

Everything above is **prediction and correlation**. Nothing in this report establishes that intervening on a flagged account would have prevented its cancellation.

Specifically:

- **The reaction-time findings are correlational.** Accounts that received early contact may differ systematically from those that did not, being larger, better covered or more visible. That selection effect is not controlled here.
- **Flagged-and-saved is not evidence of a save.** Some flagged accounts would have renewed regardless. Without a control arm, that fraction is unknown.
- **Historical accuracy is not a forward guarantee.** It measures a model against a period that has already happened, under conditions that may not persist.

Establishing causation requires a randomised holdout running forward: a stratified share of flagged accounts deliberately withheld from outreach, assigned before outcomes are known. That is the only design that answers "did the intervention work," and this report is not a substitute for it.

8 SEALED FORWARD FORECAST

The forecast below was committed before the period began, and its hash was published at the time of issue. It cannot be revised after the fact. It will be scored against outcomes on **31 December 2026** and the result published whether or not it is favourable.

COMMITMENT	FORECAST	INTERVAL
Churned ARR, Q3–Q4 2026	\$1.83M	\$1.55M – \$2.14M
Gross revenue retention	85.9%	84.1% – 87.4%
Share flaggable at 90 days	37%	31% – 43%
Accounts named in the sealed at-risk list	214	,

Commitment hash (SHA-256):

4f2a9c7e18b6d035ae91cf4d7b2e6083c5194ad7fe28b0c6193d7a54eb0f2c88

Verification: the sealed list and its hash were recorded on issue. Re-hashing the delivered list must reproduce the value above. A mismatch means the forecast was altered and the result should be discarded.

Method. Account state was reconstructed to its as-of value at each prediction date from an append-only event ledger. Labels were derived from billing and subscription events, never from present-day account status. Evaluation is walk-forward with out-of-time splits, a 14-day embargo, and isotonic calibration fitted within each training window only.

Sample report · synthetic figures · prepared 8 August 2026 · Indextxero